

ASSIGNMENT 1-BASED ON SDG GOALS

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Analytical Assignment

Topic:

Explain how Artificial Intelligence assists doctors in medical image analysis.

Related SDG

SDG 3 – Good Health and Well-being

1. Problem Understanding

Medical image analysis helps doctors diagnose diseases using X-rays, CT scans, MRI scans, and ultrasound images. Manual analysis is time-consuming and may lead to errors. Artificial Intelligence (AI) can analyse medical images quickly and accurately, helping doctors detect diseases at an early stage and improve patient care.

2. Objectives

- To understand the role of AI in medical image analysis.
- To detect diseases from medical images accurately.
- To reduce diagnosis time.
- To assist doctors in making better clinical decisions.
- To improve healthcare quality.

3. Data Collection and Preparation :

Dataset:

Chest X-ray Images Dataset (or any medical image dataset from Kaggle)

Data Collection:

- Download the medical image dataset from Kaggle.
- The dataset contains Normal and Pneumonia chest X-ray images.

Data Preparation:

- Remove duplicate and corrupted images.
- Resize all images to 224×224 pixels.
- Normalize pixel values to improve model performance.
- Label the images as Normal (0) and Pneumonia (1).
- Split the dataset into 80% training and 20% testing.

4. Analysis Using Machine Learning

AI Techniques Used

- Convolutional Neural Network (CNN)
- Deep Learning
- Image Classification

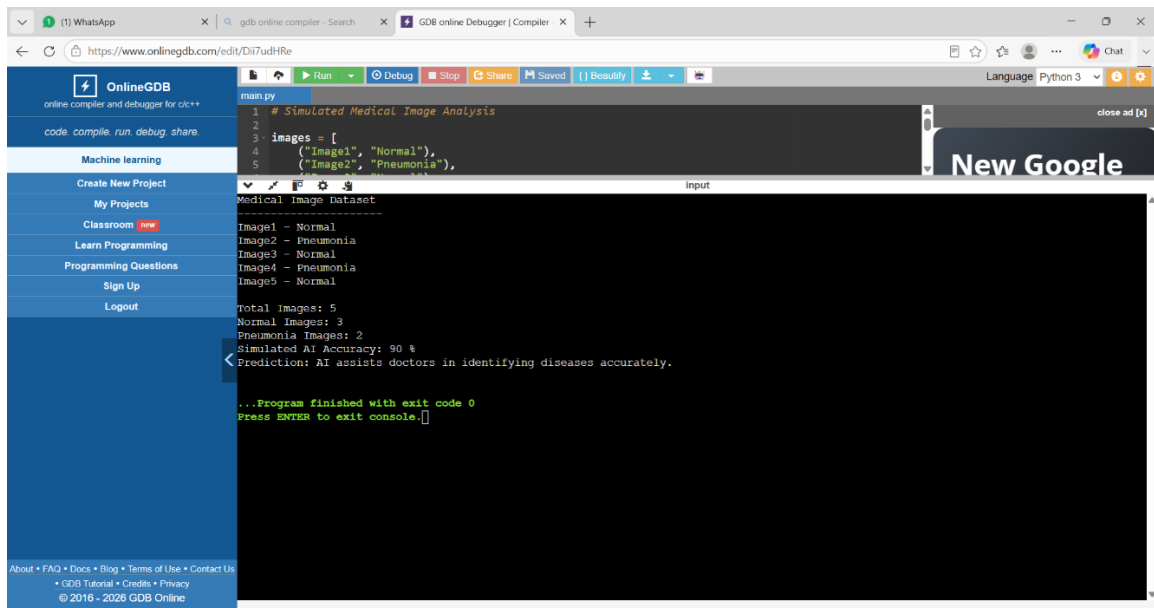
Working

- Input medical image.
- Preprocess the image.
- Extract image features.
- Classify as Normal or Disease.
- Display prediction to the doctor.

Simulated code :

```
# Simulated Medical Image Analysis
images = [
    ("Image1", "Normal"),
    ("Image2", "Pneumonia"),
    ("Image3", "Normal"),
    ("Image4", "Pneumonia"),
    ("Image5", "Normal")
]
print("Medical Image Dataset")
print("-----")
normal = 0
pneumonia = 0
for image, diagnosis in images:
    print(image, "-", diagnosis)
    if diagnosis == "Normal":
        normal += 1
    else:
        pneumonia += 1
print("\nTotal Images:", len(images))
print("Normal Images:", normal)
print("Pneumonia Images:", pneumonia)
accuracy = 90
print("Simulated AI Accuracy:", accuracy, "%")
if accuracy >= 85:
    print("Prediction: AI assists doctors in identifying diseases accurately.")
else:
    print("Prediction: Model needs improvement.")
```

Output :



```
main.py
1 # Simulated Medical Image Analysis
2
3 images = [
4     ("Image1", "Normal"),
5     ("Image2", "Pneumonia"),
6     ("Image3", "Normal"),
7     ("Image4", "Pneumonia"),
8     ("Image5", "Normal")
9 ]
10
11 # Simulated AI Model Results
12 Medical Image Dataset
13 -----
14 Image1 - Normal
15 Image2 - Pneumonia
16 Image3 - Normal
17 Image4 - Pneumonia
18 Image5 - Normal
19
20 Total Images: 5
21 Normal Images: 3
22 Pneumonia Images: 2
23 Simulated AI Accuracy: 90 %
24 Prediction: AI assists doctors in identifying diseases accurately.
25
26 ...Program finished with exit code 0
27 Press ENTER to exit console.
```

5. Visualization and Interpretation

Possible visualizations include:

- Bar chart showing the number of normal and diseased images.
- Pie chart showing disease distribution.
- Accuracy graph of the AI model.
- Confusion Matrix.

6. Innovation and SDG Mapping

SDG 3 – Good Health and Well-being

AI supports SDG 3 by:

- Early disease detection.
- Improving diagnostic accuracy.
- Reducing human errors.
- Saving doctors' time.
- Providing faster treatment decisions.
- Improving patient outcomes.

7. Advantages

- High accuracy.
- Faster diagnosis.
- Early disease detection.
- Supports doctors in decision-making.
- Reduces workload.

8. Limitations

- Requires a large amount of medical data.
- High computational cost.
- Data privacy concerns.
- Predictions depend on data quality.

9. Conclusion

Artificial Intelligence plays an important role in medical image analysis by helping doctors detect diseases quickly and accurately. AI reduces diagnostic errors, saves time, and improves healthcare services. It supports SDG 3 (Good Health and Well-being) by enabling early diagnosis and better patient care.

10. References

1. [Kaggle – Chest Radiograph Images (Pneumonia & Normal) Dataset](https://www.kaggle.com/datasets/sangeethakallat/chest-radiograph-images-pneumonia-and-normal?utm_source=chatgpt.com)
2. [Kaggle – NIH Chest X-ray Dataset](https://www.kaggle.com/datasets/nih-chest-xrays/data?utm_source=chatgpt.com)
3. Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-Scale Chest X-Ray Database and Benchmarks on Weakly-Supervised Classification and Localization of Common Thorax Diseases. IEEE CVPR.
4. [Kaggle – Chest X-Ray Images (Pneumonia) Dataset](https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia?utm_source=chatgpt.com)